

Labwork 2

A1/ Value of middle pixel:

- Average filter: 28, 33

- Weighted average filter: 29, 37.5

The second filter gives more importance to the middle pixel, since it has a higher weight than the rest of the pixels.

A2/ Sorted pixels: 12, 12, 13, 13, 14, 14, 15, 16, 255

→ new middle pixel: 14

The median filter prevents anomalies (unusually high/low pixels) from affecting the rest of the pixel neighborhood.

A3/ Output at center: 320

This mask outputs high value if center pixel is brighter than the average of the rest of the pixels.

$$A4/ \left. \begin{array}{l} G_x = 0 \\ G_y = 360 \end{array} \right\} G = 360$$

The main edge is horizontal, since G_y (detects vertical changes) $\gg G_x$ (detects horizontal changes)

A5/ Laplacian at center: $(10 \cdot 4 \cdot 1) + 50 \cdot (-4) = -160$

$$\rightarrow \text{center pixel: } 50 - (-160) = 210$$